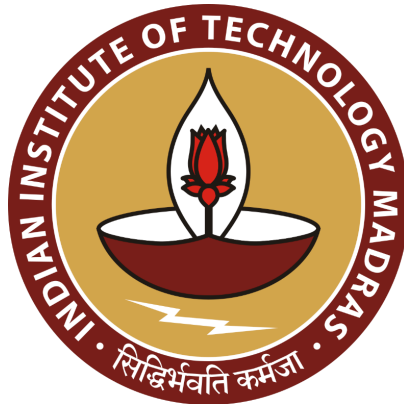


Risk-Based Prioritisation for Misinformation Review

Industry 4.0 Project Report

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Team 7- GAABS 4.0



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1. Problem Identification: Objective, Constraints, and Decision Variables

The Problem: The rapid spread of online misinformation poses a significant societal threat. While manual fact-checking by human reviewers is a critical line of defense, it is a scarce and expensive resource. Social media platforms generate far more content than can be feasibly reviewed. A random or "first-in, first-out" review queue is highly inefficient, allowing high-harm content to achieve significant viral reach before it is ever flagged.

Objective: The primary objective is to maximise the expected harmful engagement intercepted by manual review. This moves beyond simple classification (True/False) and prioritises content that is both likely to be false and likely to be seen by many people.

Formally, the objective is:

$$\text{Maximise } \sum_i x_i \cdot (p_i \times \text{engagement}_i)$$

Where:

- i is an individual post.
- x_i is the decision variable (1 if reviewed, 0 if not).
- p_i is the predicted probability that post i is misinformation.
- engagement_i is the expected engagement (reach) of post i .

Decision Variables: x_i in $\{0, 1\}$: A binary decision variable for each post i , indicating whether it should be routed to the human fact-checking queue (1) or not (0).

Constraints: The system operates under a primary resource limitation where the total number of posts reviewed per day cannot exceed the available reviewer capacity, K .

2. Relevant Data

Source: [Gen AI Misinformation Detection Dataset 2024-2025](#)

This dataset captures realistic simulations of news articles and social media posts circulating between 2024–2025, labeled for potential AI-generated misinformation.

It includes 500 rows $\times$ 31 columns, combining:

- **Temporal features** $\rightarrow$ date, time, month, day of week

- **Text-based metadata** → platform, region, language, topic
- **Quantitative engagement metrics** → likes, shares, comments, CTR, views
- **Content quality indicators** → sentiment polarity, toxicity score, readability index
- **Fact-checking signals** → credibility source score, manual check flag, claim verification status
- **Target variable** → is_misinformation (0 = authentic, 1 = misinformation)

3. Approach to Solve the Problem

A three-stage analytics methodology was employed to move from data to a prescriptive operational solution.

1. Exploratory Analytics: First, the dataset was analyzed to map risk hotspots and validate feature utility. This involved creating visualisations (e.g., heatmaps, bar charts) to understand which platforms, countries, and times of day had the highest prevalence of misinformation. This step confirmed that risk is highly concentrated, justifying a targeted approach over a random one.

2. Predictive Modelling: Next, a predictive model was built to estimate the probability (p_i) that any given post is misinformation.

- **Pipeline:** The text of each post was processed using TF-IDF, and these text features were combined with the scaled metadata and categorical features (e.g., platform, country).
- **Model:** A Random Forest classifier was trained, proving effective at identifying drivers. The model's feature importance (see Figure 86 in the report) confirmed that token_count, sentiment_score, text_length, toxicity_score, and engagement were all top predictors.

3. Prescriptive Optimisation: Finally, the model's predictive output (p_i) was used to create a prescriptive solution.

- **Risk Score:** Instead of just using the probability, we calculated an Expected Harm score for each post:

$$Harm = p_i \times engagement_i$$
- **Prioritisation:** All incoming posts are ranked by this Harm score in descending order.
- **Action:** The top K posts from this ranked list are routed to the human review queue, aligning with the capacity constraint.

Results & Implications: This risk-based prioritisation method dramatically outperforms the random review baseline. The Coverage Curve (Figure 1) shows that by reviewing only the top 30% of risk-prioritised posts, we successfully intercept and capture approximately 56% of the total harmful engagement. This represents a ~1.87x efficiency gain, effectively doubling the impact of the limited fact-checking team.

Repository with the code files can be found here: github.com/ALikesToCode/industry-4.0-gaabs-analysis/